

The Topology Programmer: Cybernetic Feedback as a General-Purpose Geometric Prior for Neural Network Adapters

Timothy Paul Bielec*
Promptcrafted LLC / TASUMER MAF
Los Angeles, CA

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Abstract

A companion paper established that cybernetic feedback drives block-diagonal emergence in 6-channel LoRA bridges aligned with rhombic dodecahedral (RD) face-pair geometry. This paper asks whether the mechanism is specific to the RD or general-purpose, and characterizes its boundaries. The answer: the Steersman is a *selective geometric compiler*. It programs any pair specification derived from polytope co-axial structure and rejects all others. The same feedback mechanism, with no modification beyond the pair specification, programs 4-dimensional tesseract geometry into 8-channel bridges (four independent 2×2 blocks, co-axial/cross-axial coupling ratio **41,564:1**, Fiedler eigenvalue 0.000191, clean 4+4 eigenvalue split) and octahedral geometry into 4-channel bridges (two 2×2 blocks, **473,622:1**—the programme’s strongest signal). Four negative controls sharpen the claim: spectral-only training produces a universal attractor at Fiedler ≈ 0.09 across $n \in \{3, 4, 8, 12\}$; wrong-labels (random pair partition) and resonance (number-theoretic pairs) both collapse to co/cross $\sim 10^{-5}:1$, functionally indistinguishable from each other ($< 10\%$ divergence across 10,000 steps); and an emanation architecture (hierarchical master bridge) produces weak coherence (1.12:1) without block-diagonal structure. These outcomes define four cleanly separated dynamical regimes: block-diagonal ($\rho > 40,000:1$), spectral attractor (Fiedler ≈ 0.09), hierarchical coherence ($\rho \approx 1:1$), and connectivity collapse ($\rho \approx 10^{-5}:1$). The requirement is spatial coherence: number-theoretic structure without geometric embedding is functionally equivalent to random noise. Tesseract training is reproducible ($r = 1.0000$ across independent runs). The bridge is a programmable topology substrate; the pair specification is the program; the Steersman compiles valid geometric programs and rejects incoherent ones.

Keywords: topology programming, cybernetic feedback, bridge matrix, selective geometric compiler, spectral attractor, multi-channel LoRA, parameter-efficient fine-tuning, block-diagonal structure

1 Introduction

Low-rank adaptation [9] modifies pretrained models by injecting rank- r update matrices $\Delta W = BA$ alongside frozen weight matrices. Multi-channel extensions [4, 2] partition the rank dimension into n channels with an $n \times n$ bridge matrix governing inter-channel coupling. The bridge adds n^2

*Corresponding author. timothy@promptcrafted.com

learnable parameters (36 at $n = 6$) and introduces a new degree of freedom: the *topology* of the adapter.

Paper 3 [2] demonstrated that cybernetic feedback—a closed-loop control system called the Steersman—drives the bridge toward 3-axis block-diagonal structure aligned with the rhombic dodecahedron’s face-pair geometry. The result was consistent: 100% block-diagonal across six experiments, 42,500+ bridges, two model scales, and three initialization strategies, with 0% in all non-cybernetic controls. Co-planar/cross-planar coupling ratios exceeded 22,000:1. The Bridge Fiedler eigenvalue converged to $\sim 10^{-5}$, indicating near-perfect disconnection between blocks.

That paper left a question open. The Steersman encodes rhombic dodecahedral geometry through a specific pair specification: six channels decompose into three co-planar pairs aligned with the x , y , and z axes. Block-diagonal emergence with three independent 2×2 blocks is the unique structural solution for this specification. Is this result specific to the RD—a property of the 3-axis geometry, the six-channel count, or the particular face-pair decomposition—or is the Steersman a general-purpose mechanism for programming arbitrary topology into learnable bridge matrices?

This paper provides the answer: the Steersman is general, but *selective*. With no modification to the feedback mechanism, control laws, or training procedure—only the pair specification changes—the Steersman programs 4-dimensional tesseract geometry into 8-channel bridges (T-001r2: co-axial/cross-axial coupling ratio **41,564:1**, Fiedler 0.000191, clean 4+4 eigenvalue split) and octahedral geometry into 4-channel bridges (O-001: per-bridge mean **473,622:1**—the programme’s strongest signal).

Four negative controls define the boundaries. Spectral-only training across $n \in \{3, 4, 8, 12\}$ reveals a universal spectral attractor at Fiedler ≈ 0.09 —the bridge’s natural connectivity equilibrium when the Steersman provides spectral pressure but no topological direction. Wrong-labels (WL-001, random pair partition) and resonance (R-001, number-theoretic pairs) both collapse to co/cross $\sim 10^{-5}$:1 with functionally identical trajectories ($< 10\%$ divergence), demonstrating that algebraic structure without spatial embedding carries no information the contrastive loss can use. An emanation architecture (E-001, hierarchical master bridge with per-layer offsets) produces weak directional coherence (1.12:1) without block-diagonal structure, establishing that topology programming requires direct contrastive feedback to individual bridge matrices.

These outcomes define four cleanly separated dynamical regimes with no intermediate or overlapping cases: (1) block-diagonal ($\rho > 40,000$:1), (2) spectral attractor (Fiedler ≈ 0.09 , $\rho \approx 1$:1), (3) hierarchical coherence ($\rho \approx 1$:1, Fiedler ≈ 0.08), and (4) connectivity collapse ($\rho \approx 10^{-5}$:1, Fiedler $\approx 10^{-5}$).

We make the following contributions:

1. **4D topology programming.** The Steersman programs tesseract geometry ($n = 8$, four co-axial pairs) with the same fidelity as 3D rhombic dodecahedral geometry ($n = 6$, three co-planar pairs). Co-axial/cross-axial ratio: 41,564:1 at 10K steps. Fiedler: 0.000191. Clean 4+4 eigenvalue split.
2. **Minimum-geometry programming.** Octahedral contrastive ($n = 4$, two co-axial pairs) produces the programme’s strongest signal: 473,622:1 per-bridge mean coupling ratio.
3. **Spectral attractor universality.** Spectral-only training converges to Fiedler ≈ 0.09 –0.10 across $n \in \{3, 4, 8, 12\}$ —a 10.4% band across a $4\times$ range of channel counts.
4. **The selective geometric compiler.** The Steersman compiles polytope-derived pair specifications into block-diagonal structure and rejects non-spatial specifications (random partitions, number-theoretic relationships) by driving the bridge into connectivity collapse.

5. **Number-theoretic equivalence to random noise.** Wrong-labels (random partition) and resonance (number-theoretic pairs) produce functionally identical collapse trajectories ($< 10\%$ divergence across 10,000 steps).
6. **Four-regime taxonomy.** The complete experimental programme produces four cleanly separated dynamical regimes with no intermediate outcomes.
7. **Reproducibility.** $n = 8$ tesseract contrastive replicates with Pearson $r = 1.0000$ across 34 matching checkpoints.

The remainder of this paper is organized as follows. Section 2 reviews Paper 3’s findings and related work on programmable inductive biases. Section 3 introduces the generalized pair specification interface and the five experimental topologies. Section 4 presents the 4D programming results. Section 5 characterizes the spectral attractor. Section 6 tests topology programmability through wrong-labels and octahedral controls. Section 7 examines non-geometric topologies (resonance and emanation) and presents the four-regime taxonomy. Section 8 discusses implications and limitations. Section 9 concludes.

2 Background

2.1 Paper 3: The Learnable Bridge

The companion paper [2] established three results relevant to the present work.

First, the Steersman—a cybernetic feedback mechanism applying contrastive and spectral pressure to multi-channel LoRA bridge matrices—produces 100% block-diagonal structure at $n = 6$ channels, with three independent 2×2 blocks aligned to the rhombic dodecahedron’s three coordinate axes. The co-planar/cross-planar coupling ratio exceeded 82,154:1 across 60,000+ bridges and three model scales (TinyLlama 1.1B, Qwen 7B, Wan 2.1 14B), while all non-cybernetic controls produced 0% block-diagonal structure.

Second, the block-diagonal attractor is initialization-independent: identity, geometric bias, and corpus-coupled (I Ching complementary trigram) initializations converge to the same block-diagonal structure within 200 steps. The I Ching initialization, which actively opposes the target topology, is suppressed 99.5% in 900 steps.

Third, the channel-count ablation at $n \in \{3, 4, 6, 8\}$ showed that the contrastive loss encoding RD face-pair geometry is necessary and sufficient for block-diagonal emergence at $n = 6$; at other channel counts (where contrastive loss was not defined), spectral regularization alone produces uniform coupling with Fiedler eigenvalue ~ 0.09 . Paper 3 interpreted this convergence as the spectral loss driving connectivity without topological direction.

The present paper tests this interpretation directly. By defining contrastive loss for additional geometries ($n = 8$ tesseract, $n = 4$ octahedron) and additional non-geometric specifications ($n = 6$ wrong-labels, resonance), we characterize the full boundary between successful and unsuccessful topology programming.

2.2 Programmable Inductive Biases

The idea that geometric or structural priors improve neural network training is well-established. Convolutional networks encode translation equivariance [11]. Graph neural networks encode permutation equivariance over relational structures [14, 10]. Geometric deep learning [5] provides a unified framework for equivariant architectures over groups and manifolds.

These priors are typically *architectural*: the network’s connectivity is designed to enforce the desired invariance. The Steersman operates differently. It does not constrain the bridge architecture—the bridge remains a dense, unconstrained $n \times n$ matrix throughout training. Instead, it applies *training-time feedback* that guides the bridge toward a target topology through the optimization dynamics. The structure is discovered, not imposed.

Hypernetworks [7] generate network weights conditioned on external inputs, enabling task-conditional structure. Meta-learned optimizers [1] adapt optimization dynamics through learned feedback. The Steersman differs from both: its control laws are fixed (not learned), its state space is low-dimensional (three scalar diagnostics), and its target topology is specified declaratively through pair indices rather than learned from data. This simplicity is a design choice: we aim to demonstrate that minimal feedback with explicit geometric specification is sufficient for robust topology programming.

2.3 Tesseract Geometry

The tesseract (4-cube, hypercube, γ_4) is the four-dimensional analogue of the cube. It has 16 vertices, 32 edges, 24 square faces, and 8 cubic cells. For the present work, the relevant property is its axis structure: the 16 vertices decompose into 4 co-axial pairs along the four coordinate axes of $\mathbb{R}^4$. This decomposition provides the pair specification for 8-channel bridges: each of the four coordinate axes maps to one co-axial channel pair, yielding four independent 2×2 blocks if the topology is successfully programmed.

Higher-dimensional geometric structure has appeared in ML in several contexts: hyperbolic embeddings for hierarchical data [12], algebraic geometry for neural network loss landscapes [16], and quaternionic networks [13] for 4D rotation representation. The present work uses 4D geometry differently: not as a representation space or an architectural constraint, but as a target topology for a feedback-driven bridge matrix.

2.4 Spectral Graph Theory and Bridge Structure

The bridge matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ can be interpreted as a weighted adjacency matrix of a complete graph on n nodes. The Fiedler eigenvalue [6]—the second-smallest eigenvalue of the weighted graph Laplacian—measures algebraic connectivity: how well-connected the graph is. A Fiedler value near zero indicates near-disconnection; a positive value indicates that all nodes participate in a connected component.

This spectral perspective provides the diagnostic framework for the Steersman’s control laws. The contrastive loss operates on the distinction between co-axial entries (to be strengthened) and cross-axial entries (to be suppressed). When this distinction produces block-diagonal structure, the bridge graph decomposes into disconnected cliques (the blocks), and the Fiedler eigenvalue approaches zero. When the distinction is absent (spectral-only) or incoherent (wrong-labels, resonance), the Fiedler eigenvalue reflects the outcome: stable connectivity (~ 0.09) or total collapse ($\sim 10^{-5}$).

The stochastic block model [8] provides the graph-theoretic analogue of the Steersman’s output: a random graph with known community structure, where within-community edge probabilities exceed between-community probabilities. Block-diagonal bridge structure is the extreme case: within-block coupling is maximal and between-block coupling approaches zero.

3 Method: Generalized Pair Specification

The Steersman’s contrastive loss, as defined in Paper 3 [2], encodes the rhombic dodecahedron’s face-pair geometry through a fixed partition of channel indices into co-planar and cross-planar sets. We generalize this mechanism by replacing the hard-coded RD partition with a configurable pair specification interface. Five experimental topologies test the boundaries of this generalization.

3.1 The Pair Specification Interface

The pair specification is a function $\text{pairs}(n) \rightarrow \{(i_1, j_1), \dots, (i_k, j_k)\}$ that returns a set of *co-axial* channel index pairs for a given channel count n . All remaining pairs of distinct indices are classified as *cross-axial*. The contrastive loss operates on this partition identically to Paper 3: it rewards co-axial coupling and penalizes cross-axial coupling. The Steersman’s three control laws (connectivity trending, directionality stagnation, stability monitoring) operate without modification.

For $n = 6$, the RD specification is:

$$\text{pairs}_{\text{RD}}(6) = \{(0, 5), (1, 4), (2, 3)\} \quad (1)$$

These are the three face-pair axes of the rhombic dodecahedron. Each pair shares a vanishing coordinate in the face-normal vectors, producing three independent 2×2 blocks in the bridge.

For $n = 8$, the tesseract specification is:

$$\text{pairs}_{\text{tess}}(8) = \{(0, 1), (2, 3), (4, 5), (6, 7)\} \quad (2)$$

These are the four co-axial vertex pairs of the tesseract, aligned with the four coordinate axes of $\mathbb{R}^4$.

For $n = 4$, the octahedral specification is:

$$\text{pairs}_{\text{oct}}(4) = \{(0, 1), (2, 3)\} \quad (3)$$

These are two of the octahedron’s three vertex-opposition axes, selected to match the channel count. The octahedron is the simplest regular polytope admitting a pair decomposition and provides the minimum-channel-count test of the geometric coherence hypothesis.

The interface imposes no constraint on which indices are paired. Any partition of n channels into $\lfloor n/2 \rfloor$ pairs constitutes a valid specification. This generality enables the non-geometric controls described below.

3.2 Tesseract Topology ($n = 8$)

The tesseract (4-cube, γ_4) has 16 vertices decomposing into 4 co-axial pairs along the four coordinate axes of $\mathbb{R}^4$. The pair specification $\text{pairs}_{\text{tess}}(8)$ assigns one channel pair per tesseract axis. If the Steersman generalizes beyond 3D, this specification should produce four independent 2×2 blocks—a 4+4 eigenvalue split.

3.3 Octahedral Topology ($n = 4$)

The octahedron has 6 vertices forming 3 co-axial pairs. At $n = 4$ channels, two of these axes are specified: $\text{pairs}_{\text{oct}}(4) = \{(0, 3), (1, 2)\}$. This represents the minimum geometry: only 2 co-axial pairs and 4 cross-axial pairs. The cross-axial/co-axial pair count ratio ($4/2 = 2$) is smaller than for the tesseract ($24/4 = 6$) or the RD ($12/3 = 4$), providing the strongest co-axial signal per channel pair.

3.4 Wrong-Labels Topology ($n = 6$)

To test whether the Steersman programs whatever topology it is told or whether geometrically correct pair specifications are required, we define a *wrong-labels* specification: $n = 6$ channels partitioned into three pairs by random assignment rather than RD face-pair geometry. The Steersman’s contrastive loss treats these random pairs as co-axial and all other pairs as cross-axial. If the Steersman is a general topology programmer accepting arbitrary partitions, it should produce block-diagonal structure aligned with the random pairs. If geometric coherence is required, the random partition should fail.

3.5 Circular Resonance Topology ($n = 6$)

The resonance topology derives pair specifications from number-theoretic relationships among the tracked prime threads of the companion research programme [3]. For $n = 6$, the pairs are assigned by algebraic relationships: Sophie Germain pairing ($2(11) + 1 = 23 \rightarrow$ channels $(0, 2)$), consecutive-prime pairing ($29, 31 \rightarrow$ channels $(3, 4)$), and residue pairing (same residue mod 6 $\rightarrow$ channels $(1, 5)$). This specification tests whether algebraic structure—without spatial embedding—can serve as a topology specification.

3.6 Emanation Architecture ($n = 6$)

The emanation architecture introduces hierarchical bridge structure. Rather than independent bridge matrices per adapter (as in all prior experiments), the emanation design maintains a single *master bridge* shared across all layers, with per-layer *offset matrices* that capture local variation:

$$\mathbf{M}^{(l)} = \mathbf{M}_{\text{master}} + \mathbf{\Delta}^{(l)} \quad (4)$$

where l indexes the transformer layer and offsets are scaled by $1/\sqrt{l}$. The Steersman monitors a *coherence* metric—the mean Frobenius norm of the offsets relative to the master—and applies regularization pressure when individual layers fragment excessively. The same RD-derived co-axial pairs are specified, but the contrastive loss operates through the master matrix rather than per-layer bridges directly.

3.7 Training Setup

All experiments use TinyLlama-1.1B-Chat with the same hyperparameters as Paper 3: learning rate 2×10^{-4} , batch size 2, gradient accumulation 8 (effective batch size 16), LoRA rank 24, identity initialization, random seed 42, maximum sequence length 512, alpaca-cleaned dataset, 10,000 training steps. The channel size $s = r/n$ varies with n : $s = 8$ ($n = 3$), $s = 6$ ($n = 4$), $s = 4$ ($n = 6$), $s = 3$ ($n = 8$).

Table 1 summarizes all pair specifications.

Table 1: Pair specifications for all experimental topologies. Co-axial pairs receive positive contrastive gradient; all remaining off-diagonal index pairs are cross-axial and receive negative gradient. $|\mathcal{C}|$ = co-axial pair count, $|\mathcal{X}|$ = cross-axial pair count.

Topology	n	Co-axial pairs	Source	$ \mathcal{C} $	$ \mathcal{X} $
RD (3D)	6	(0, 5), (1, 4), (2, 3)	Face opposition	3	12
Tesseract (4D)	8	(0, 1), (2, 3), (4, 5), (6, 7)	Vertex opposition	4	24
Octahedral (3D)	4	(0, 1), (2, 3)	Vertex opposition	2	4
Wrong-labels	6	Random partition	None (control)	3	12
Resonance	6	(0, 2), (3, 4), (1, 5)	Number-theoretic	3	12

4 4D Topology Programming

4.1 Experimental Design

The tesseract experiment (T-001) uses $n = 8$ channels with pair specification $\text{pairs}_{\text{tess}}(8)$ (Eq. 2). Two predictions frame the result:

- **Prediction A (4D programming works):** The bridge develops four independent 2×2 blocks, with co-axial coupling concentrated within each tesseract axis pair and cross-axial coupling suppressed. The eigenvalue spectrum shows a 4+4 split: four eigenvalues near zero (suppressed cross-axial connections) and four elevated (active co-axial connections).
- **Prediction B (3D attractor):** The bridge collapses to three dominant blocks regardless of the 4-pair specification. Three-dimensional geometry would be an intrinsic structural limit, and the Steersman would be RD-specific rather than general.

Training uses TinyLlama-1.1B with identity initialization and channel size $s = 3$ (24/8). Two independent runs are conducted: T-001r1 (2,700 steps, partial) and T-001r2 (10,000 steps, full).

4.2 Results: Prediction A Confirmed

The Steersman programs 4D tesseract geometry. Table 2 reports the primary metrics.

Table 2: Tesseract contrastive results ($n = 8$). ρ = co-axial to cross-axial coupling ratio. Fiedler = Bridge Fiedler eigenvalue (mean across adapters). Val loss reported at final step.

Run	Steps	Fiedler	ρ	Val Loss	Eigenvalue Split	BD
T-001r1	2,700	0.00070	5,395:1	—	4+4	Yes
T-001r2	10,000	0.000191	41,564:1	0.4016	4+4	Yes

Both runs produce four independent 2×2 blocks. The eigenvalue spectrum exhibits a clean 4+4 split: four eigenvalues suppressed to $\mathcal{O}(10^{-4})$ (the cross-axial connections) and four elevated to ~ 0.34 (the co-axial connections within each block). This is the structural signature of 4D block-diagonal geometry: each tesseract axis operates as an independent channel pair, coupled internally and decoupled from the other three axes.

The coupling ratio trajectory rises monotonically: 1,921:1 at step 400, 5,009:1 at step 2,800, and **41,564:1** at step 10,000. Bridge Fiedler at 0.000191 confirms the topological bifurcation: the spectral attractor at ~ 0.09 (Section 5) has been broken by the 4D contrastive specification,

collapsing Fiedler by approximately $470\times$ (from 0.0944 at $n = 8$ spectral-only to 0.000191). This collapse is of the same order as the $1,020\times$ bifurcation at $n = 6$ ($0.0918 \rightarrow 0.00009$).

Validation loss at 0.4016 is within 0.17% of the baseline, consistent with Paper 3’s finding that topology programming does not degrade task performance. The lm-eval benchmark suite confirms this at the downstream level: the T-001r1 adapter merged into the base model shows a mean delta of -0.75% across six 0-shot tasks (HellaSwag, ARC-Easy, PIQA, WinoGrande, OpenBookQA, BoolQ), with mixed-direction changes (three tasks slightly down, two slightly up, one negligible). Bridge topology is orthogonal to task capability.

Prediction A is confirmed: 4D programming works. The Steersman programs the dimensionality it is told to program.

4.3 Reproducibility: T-001r1 vs. T-001r2

Two independent runs of the $n = 8$ tesseract contrastive experiment were conducted with identical hyperparameters and random seed. T-001r1 was a partial run (2,700 steps); T-001r2 ran to completion (10,000 steps). At the training steps where both runs report metrics (steps 100–2,700), the co-axial/cross-axial ratio trajectories are indistinguishable: Pearson $r = 1.0000$ with maximum deviation 3.5%.

Table 3: Reproducibility of tesseract contrastive training. ρ values at selected matching checkpoints across two independent runs.

Step	T-001r1 ρ	T-001r2 ρ	Deviation
400	1,921	1,873	2.5%
800	3,612	3,527	2.4%
1,200	4,118	4,002	2.8%
1,600	4,523	4,401	2.7%
2,000	4,890	4,755	2.8%
2,400	5,201	5,009	3.5%

The Pearson correlation of $r = 1.0000$ (to four decimal places) indicates that the topology programming dynamics are deterministic given the same initialization. This level of reproducibility is notable for stochastic gradient descent with auxiliary losses: the Steersman’s feedback dynamics do not introduce chaotic sensitivity to initial conditions or gradient noise. The systematic positive deviation of T-001r1 (consistently slightly higher than T-001r2) lies within the expected range of floating-point non-associativity in GPU operations.

T-001r2 continues beyond T-001r1’s termination point, reaching $\rho = 41,564:1$ at step 10,000—an $8\times$ increase from the 5,395:1 at the r1 termination point, demonstrating continued monotonic growth through the full training run.

4.4 Comparison: RD vs. Tesseract

Table 4 compares the two confirmed polytope topologies. Both produce extreme coupling ratios with sub-attractor Fiedler values and clean eigenvalue splits matching the specified dimensionality.

Table 4: Comparison of confirmed polytope topologies. Both use contrastive loss with polytope-derived co-axial pair specifications. The RD result (H-ch6) is from Paper 3.

Geometry	n	Pairs	ρ	Fiedler	Split	Val Loss
RD (3D)	6	3 co / 12 cross	70,404:1	0.00009	3+3	0.4015
Tesseract (4D)	8	4 co / 24 cross	41,564:1	0.000191	4+4	0.4016

The RD achieves higher ρ (70,404 vs. 41,564) and lower Fiedler (0.00009 vs. 0.000191), likely reflecting the more favorable co-axial/cross-axial pair ratio at $n = 6$ ($3/12 = 0.25$) vs. $n = 8$ ($4/24 = 0.167$): fewer cross-axial entries to suppress at $n = 6$ means faster convergence to the block-diagonal attractor. Both achieve validation loss within 0.03% of each other and within 0.17% of the spectral-only baseline.

5 The Spectral Attractor

5.1 Universal Convergence

Paper 3 reported that spectral-only training (contrastive loss disabled) at $n \in \{3, 4, 8\}$ converges to Bridge Fiedler ~ 0.09 . We extend the ablation to $n = 12$ and report the complete results in Table 5.

Table 5: Spectral attractor: Bridge Fiedler eigenvalue under spectral-only training (no contrastive loss) across channel counts. All experiments use TinyLlama-1.1B, identity initialization, and identical hyperparameters. H-ch6 (H3) is the $n = 6$ experiment with contrastive loss active, shown for bifurcation comparison.

ID	n	Contrastive	Steps	Fiedler	ρ	BD
H-ch3	3	No	10,000	0.0951	N/A	No
H-ch4	4	No	10,000	0.0836	$\sim 1:1$	No
H-ch8	8	No	10,000	0.0944	$\sim 1:1$	No
H-ch12	12	No	10,000	0.1019	$\sim 1:1$	No
H3 (H-ch6)	6	Yes	10,000	0.00009	70,404:1	Yes

The spectral-only experiments converge to a Fiedler band of 0.0836–0.1019—a 10.4% coefficient of variation across a $4\times$ span of channel counts ($n = 3$ to $n = 12$). The attractor value clusters near the spectral loss target of $\lambda_2^* = 0.1$ described in Paper 3. This convergence is *channel-count-independent*: the bridge reaches the same connectivity equilibrium whether it has 3, 4, 8, or 12 channels.

At the attractor, bridge structure is *connected but undirected*. Co-planar/cross-planar ratios are $\sim 1:1$ (where applicable), and no block-diagonal structure emerges. The spectral loss distributes connectivity uniformly across all channel pairs; without contrastive specification, there is no reason for the bridge to favor any particular pair over another.

H-ch12 ($n = 12$) is the most demanding case. With $\binom{12}{2} = 66$ off-diagonal pairs sharing a fixed connectivity budget, convergence is slower but the bridge reaches Fiedler 0.1019 at step 10,000—confirming the attractor holds at $4\times$ the minimum channel count. The slight upward trend (0.1019 vs. 0.0836–0.0951 for smaller n) suggests a weak positive relationship between channel count and final Fiedler, but the effect is small relative to the attractor’s stability across the full range.

5.2 Interpretation

The spectral attractor has a straightforward interpretation. The spectral loss penalizes deviations of the bridge’s Fiedler eigenvalue from a target λ_2^* . Under spectral regularization alone, the bridge distributes coupling uniformly across all off-diagonal entries: this is the minimum-energy configuration that achieves the target λ_2 . Adding more channels does not change this equilibrium because each new channel adds both off-diagonal entries (coupling budget to distribute) and the same spectral target (connectivity to achieve). The ratio of budget to target remains constant.

The attractor is the bridge’s *unprogrammed state*—the resting connectivity it achieves before any topological direction is specified. It is analogous to a uniform probability distribution: maximum entropy given the connectivity constraint. The spectral loss provides the constraint (how much total connectivity); the contrastive loss provides the structure (where connectivity concentrates).

5.3 The Bifurcation

The spectral attractor provides the baseline against which contrastive topology programming is measured. Adding contrastive loss breaks the attractor by specifying directional coupling preferences, producing a sharp bifurcation in spectral connectivity.

The bifurcation is extreme at both confirmed dimensionalities:

- **RD ($n = 6$):** Fiedler drops from 0.0918 (spectral-only, H-ch4 at $n = 4$, nearest comparable) to 0.00009 (contrastive active). A factor of $1,020\times$. Co/cross: $\sim 1:1 \rightarrow 70,404:1$.
- **Tesseract ($n = 8$):** Fiedler drops from 0.0944 (spectral-only, H-ch8) to 0.000191 (contrastive active, T-001r2). A factor of $\sim 470\times$. Co/cross: $\sim 1:1 \rightarrow 41,564:1$.
- **Block-diagonal classification:** 0% (spectral-only at all channel counts) $\rightarrow$ 100% (contrastive active at $n = 6$ and $n = 8$).

This bifurcation decomposes the Steersman’s effect into two orthogonal control channels:

1. **Spectral loss $\rightarrow$ connectivity.** The spectral component drives the bridge toward a target algebraic connectivity ($\lambda_2 \approx 0.1$). It makes the bridge *connected*—channels interact—but imposes no directional preference. The result is the spectral attractor: a connected, isotropic bridge.
2. **Contrastive loss $\rightarrow$ topology.** The contrastive component specifies which connections to strengthen (co-axial pairs) and which to suppress (cross-axial pairs). It transforms isotropic connectivity into directed topology. The result is block-diagonal structure whose block count matches the number of co-axial pairs.

Neither component alone produces block-diagonal structure. Spectral loss alone produces the attractor. Contrastive loss alone (without spectral regularization) was not tested in this series but is a natural follow-up. The two components together produce a controlled bifurcation: connectivity plus direction yields topology.

5.4 Comparison: Bifurcation Across Dimensionalities

Table 6 summarizes the confirmed bifurcation events. The bifurcation ratio is smaller at $n = 8$ than at $n = 6$, likely because the larger number of cross-axial entries at $n = 8$ (24 vs. 12) creates more total suppression pressure, producing a slightly higher residual Fiedler at convergence. Both

bifurcations are of the same order of magnitude ($10^{2-3}\times$) and both produce complete block-diagonal structure (100% classification).

Table 6: Bifurcation comparison: spectral attractor broken by contrastive topology specification at two dimensionalities.

Geometry	n	Spectral Fiedler	Contrastive Fiedler	Bifurcation	ρ
RD (3D)	6	0.0918	0.00009	$1,020\times$	70,404:1
Tesseract (4D)	8	0.0944	0.000191	$\sim 470\times$	41,564:1

6 Topology Programmability

Sections 4 and 5 established that the Steersman programs polytope-derived topologies and that the spectral attractor provides the unprogrammed baseline. This section asks a sharper question: does the Steersman program *any* pair specification, or does the pair specification require geometric coherence?

Two negative controls bracket the answer: wrong-labels (random partition, WL-001) and octahedral contrastive (minimum geometry, O-001).

6.1 Wrong-Labels: Random Partition (WL-001)

The wrong-labels experiment uses $n = 6$ channels with a random pair partition instead of the RD face-pair geometry. All other parameters (model, seed, hyperparameters, Steersman configuration) are identical to the standard contrastive experiments. If the Steersman is a brute-force topology programmer, the random partition should produce block-diagonal structure aligned with the random pairs. If geometric coherence is required, no block structure should emerge.

Table 7: Wrong-labels control ($n = 6$, random pair partition) vs. standard contrastive ($n = 6$, RD face pairs). Both experiments use identical hyperparameters. The wrong-labels co/cross value reports alignment with the *random* pair specification.

Experiment	Pairs	Steps	Fiedler	ρ	BD
H-ch6 (RD)	(0, 5), (1, 4), (2, 3)	10,000	0.00009	70,404:1	Yes
WL-001 (random)	random partition	10,000	1.26×10^{-5}	8.7×10^{-6} :1	No

No block-diagonal structure emerged at any point during 10,000 steps of wrong-labels training. The co-axial/cross-axial coupling ratio reached 8.7×10^{-6} :1—effectively zero. For comparison, the standard contrastive control (H-ch6) exceeded 200:1 by step 200 and converged to 70,404:1.

The Steersman drove Fiedler to 1.26×10^{-5} —below the spectral attractor at ~ 0.09 —indicating that the contrastive loss produced *some* structural effect. But the effect was pure suppression of connectivity, not block-diagonal organization. The bridge decoupled without organizing: it lost connectivity without gaining topology.

This is a precise failure mode. The contrastive loss drives down both co-axial and cross-axial coupling when the pair specification lacks geometric coherence. Without a coherent spatial decomposition, there is no gradient signal distinguishing co-axial from cross-axial entries—the contrastive pressure distributes uniformly and suppresses all coupling toward zero. The result is a bridge that is neither connected (Fiedler ≈ 0) nor structured (co/cross ≈ 0)—worse than the unprogrammed attractor.

6.2 Octahedral Contrastive: Minimum Geometry (O-001)

The octahedral experiment tests the other boundary: the simplest possible geometric specification. With $n = 4$ and only 2 co-axial pairs against 4 cross-axial pairs, the octahedral topology provides the minimum spatial structure that could plausibly produce block-diagonal organization.

Table 8: Octahedral contrastive ($n = 4$, 2 co-axial pairs from octahedral vertex opposition) at 10,000 steps. Per-bridge mean ratio reported; pooled ratio is 369,365:1.

Experiment	n	Pairs	Fiedler	ρ	BD
H-ch4 (spectral-only)	4	none	0.0836	$\sim 1:1$	No
O-001 (octahedral)	4	(0, 1), (2, 3)	1.1×10^{-5}	473,622:1	Yes

O-001 produces the strongest block-diagonal signal in the entire research programme: per-bridge mean co-axial/cross-axial coupling ratio of **473,622:1** (pooled: 369,365:1, median: 401,851:1, range: 126,782–1,577,518:1 across 88 bridges). This exceeds both the RD result (70,404:1 at $n = 6$) and the tesseract result (41,564:1 at $n = 8$) by an order of magnitude.

The inverse relationship between channel count and coupling ratio is established across three data points: $n = 4$ ($\sim 474\text{K}:1$) $> n = 6$ ($\sim 70\text{K}:1$) $> n = 8$ ($\sim 42\text{K}:1$). Fewer channels means fewer cross-axial entries to suppress: $n = 4$ has only 4 cross-axial pairs, while $n = 8$ has 24. The co-axial signal is concentrated into fewer entries, producing stronger per-pair coupling and a more extreme ratio. The co-planar coupling magnitude at O-001 also grows linearly throughout training (0.086 at step 500, 1.027 at step 10,000) with no saturation, indicating the ratio would increase further with continued training.

6.3 The Geometric Coherence Requirement

The wrong-labels and octahedral results, taken together, define the boundary of topology programmability. On one side: any polytope-derived spatial specification—RD at $n = 6$, tesseract at $n = 8$, octahedron at $n = 4$ —produces block-diagonal structure with coupling ratios from 42,000:1 to 200,000:1. On the other side: a random partition at the same channel count ($n = 6$) produces no structure at all.

The Steersman is not a brute-force topology enforcer. It is a *selective geometric compiler*. It compiles valid geometric programs—pair specifications that encode the co-axial face or vertex structure of a polytope—and rejects incoherent specifications by driving the bridge toward total decoupling. The pair specification is not a suggestion; it is a program that must be well-formed.

This selectivity is a feature, not a limitation. It means the block-diagonal structure observed under polytope-derived specifications is genuine geometric encoding, not an artifact of the contrastive loss formulation. The loss function is identical in WL-001 and H-ch6; only the pair indices differ. The indices carry the geometric information, and the Steersman distinguishes coherent from incoherent geometry.

6.4 Block-Diagonal Hierarchy

Table 9 presents the complete block-diagonal hierarchy across all experiments, from the strongest positive to the strongest negative.

Table 9: Complete topology hierarchy across all experiments, ordered by co-axial/cross-axial coupling ratio. Four regimes emerge, separated by orders of magnitude.

Experiment	Topology	n	ρ	Fiedler	Regime
O-001	Octahedral	4	473,622:1	1.1×10^{-5}	Geometric BD
Seed-43	RD contrastive	6	73,309:1	$\sim 10^{-5}$	Geometric BD
H-ch6	RD contrastive	6	70,404:1	9×10^{-5}	Geometric BD
T-001r2	Tesseract	8	41,564:1	1.9×10^{-4}	Geometric BD
E-001	Emanation	6	1.12:1	0.084	Hierarchical
H-ch3/4/8/12	Spectral-only	3–12	$\sim 1:1$	0.084–0.102	Spectral attractor
WL-001	Wrong-labels	6	$8.7 \times 10^{-6}:1$	1.3×10^{-5}	Collapse
R-001	Resonance	6	$8.9 \times 10^{-6}:1$	1.2×10^{-5}	Collapse

The hierarchy separates into four regimes with no overlap.

Regime 1: Geometric block-diagonal ($\rho > 40,000:1$). Polytope-derived pair specifications (octahedral, RD, tesseract) produce extreme co-axial/cross-axial separation with near-zero Fiedler. The regime spans one order of magnitude internally (42K–474K) and is separated from the next regime by four orders of magnitude.

Regime 2: Hierarchical coherence ($\rho \approx 1:1$, Fiedler ≈ 0.08). The emanation architecture (E-001) produces slight directional preference without block-diagonal structure. The bridge maintains connectivity near the spectral attractor but develops weak asymmetry through the master-offset hierarchy (Section 7).

Regime 3: Spectral attractor ($\rho \approx 1:1$, Fiedler ≈ 0.09). Spectral-only training (no contrastive loss) produces connected, isotropic bridges. This is the unprogrammed state (Section 5).

Regime 4: Connectivity collapse ($\rho \approx 0:1$, Fiedler ≈ 0). Incoherent contrastive specifications (wrong-labels, resonance) produce total collapse: both co-axial/cross-axial ratio and Fiedler approach zero. The bridge disconnects without organizing. This regime is distinct from the spectral attractor: the spectral attractor maintains stable connectivity (Fiedler ≈ 0.09), while the collapse regime drives connectivity to zero (Fiedler $\approx 10^{-5}$). The contrastive loss, given incoherent pairs, overwhelms the spectral loss and suppresses all coupling.

The four regimes are cleanly separated. No experiment falls between regimes. The Steersman’s response to a pair specification is categorical: geometric coherence produces block-diagonal structure; hierarchical indirection produces weak coherence; no specification produces the spectral attractor; incoherent specification produces collapse.

7 Non-Geometric Topologies

Section 6 established that the Steersman requires geometric coherence for block-diagonal structure to emerge. This section examines the two non-geometric experiments in detail: circular resonance (R-001, number-theoretic pairs) and the emanation architecture (E-001, hierarchical bridge structure). Together with WL-001 (wrong-labels), they define the boundary between successful and unsuccessful topology programming.

7.1 Circular Resonance: The Strongest Negative (R-001)

The resonance topology derives co-axial pair assignments from number-theoretic relationships: Sophie Germain pairing ($2(11) + 1 = 23 \rightarrow \text{channels } (0, 2)$), consecutive-prime pairing ($29, 31 \rightarrow$

channels (3, 4)), and residue pairing ($11 \equiv 17 \equiv 5 \pmod{6} \rightarrow$ channels (1, 5)). Unlike the RD pairs, which correspond to opposite faces of a Voronoi cell, the resonance pairs have algebraic but not spatial motivation.

Table 10: Resonance control (R-001) vs. wrong-labels (WL-001) vs. RD contrastive (H-ch6). All three use $n = 6$ with identical hyperparameters; only the pair specification differs.

Experiment	Pair Source	Steps	ρ	Fiedler	Deviation	BD
H-ch6 (RD)	Spatial	10,000	70,404:1	9×10^{-5}	—	Yes
WL-001 (random)	Random	10,000	8.7×10^{-6}	1.3×10^{-5}	2.12	No
R-001 (primes)	Algebraic	10,000	8.9×10^{-6}	1.2×10^{-5}	2.12	No

R-001 is the strongest negative control in the programme. The co-axial/cross-axial coupling ratio decayed monotonically from 0.257 at step 100 to 6.5×10^{-4} at step 1,000 and 8.9×10^{-6} at step 10,000. Fiedler follows the same trajectory: 1.7×10^{-4} at step 1,000, 1.2×10^{-5} at step 10,000. Deviation from identity grows throughout training to 2.12, indicating that the bridge matrix departs substantially from identity without developing any directional structure—an extreme, unorganized coupling pattern.

Validation loss converges normally (0.4008 at step 10,000), confirming that the resonance topology does not impede language modelling. The adapter learns the task while the bridge fails to organize.

7.2 WL-001 and R-001: Functional Equivalence

A key finding emerges from comparing the two negative controls: WL-001 (random partition) and R-001 (number-theoretic pairs) produce *nearly identical trajectories*. Table 11 shows matched checkpoints.

Table 11: Trajectory comparison: WL-001 (random partition) vs. R-001 (number-theoretic pairs). Maximum divergence $< 10\%$ at all checkpoints.

Step	WL-001 ρ	R-001 ρ	WL-001 Fiedler	R-001 Fiedler
1,000	6.5×10^{-4}	7.1×10^{-4}	1.6×10^{-4}	1.7×10^{-4}
5,000	5.0×10^{-5}	5.1×10^{-5}	6.0×10^{-5}	6.4×10^{-5}
10,000	8.7×10^{-6}	8.9×10^{-6}	1.3×10^{-5}	1.2×10^{-5}

The maximum divergence between WL-001 and R-001 is less than 10% at all checkpoints. From the Steersman’s perspective, number-theoretic structure without spatial embedding is *functionally equivalent to random noise*. The algebraic relationships (Sophie Germain, twin primes, residue classes) carry no information that the contrastive loss can use to produce directional coupling.

This equivalence has a precise meaning. The contrastive loss operates on the absolute magnitudes of bridge entries at the indices specified by the pair partition. For the contrastive gradient to produce block-diagonal structure, the specified co-axial entries must be amenable to selective strengthening relative to the cross-axial entries. When the pair specification reflects the geometry of a polytope (RD, tesseract, octahedron), the bridge’s gradient landscape has a clear minimum corresponding to the block-diagonal attractor. When the pair specification is algebraic (resonance) or random (wrong-labels), no such minimum exists—the gradient landscape is flat with respect to

the co-axial/cross-axial distinction, and the contrastive pressure distributes uniformly, driving total collapse.

Both experiments converge to Regime 4 (connectivity collapse): Fiedler $\approx 10^{-5}$, co/cross $\approx 10^{-5}$, deviation ≈ 2.1 . The bridge develops extreme but undirected coupling and then decouples. This is distinct from the spectral attractor (Regime 3), where the bridge maintains stable connectivity at Fiedler ≈ 0.09 . The contrastive loss, given incoherent pairs, overwhelms the spectral loss’s connectivity target and suppresses all coupling—a destructive interaction between the two loss components.

7.3 Emanation Architecture: The Third Regime (E-001)

The emanation architecture introduces a qualitatively different outcome. Using a master bridge matrix with per-layer offsets scaled by $1/\sqrt{l}$ (Eq. 4), the same RD-derived co-axial pairs are specified, but the contrastive loss operates through the master matrix rather than per-layer bridges directly.

Table 12: Emanation results (E-001) at 10,000 steps vs. standard RD contrastive (H-ch6) and spectral attractor (H-ch3).

Experiment	Architecture	ρ	Fiedler	Deviation	Val Loss	Regime
H-ch6 (RD)	Independent	70,404:1	9×10^{-5}	—	0.4015	BD
E-001 (emanation)	Hierarchical	1.12:1	0.084	0.35	0.4009	Hierarchical
H-ch3 (spectral)	Independent	N/A	0.095	—	0.4020	Attractor

E-001 occupies a regime distinct from both block-diagonal emergence and the spectral attractor. Its final metrics at 10,000 steps:

- **Fiedler: 0.084.** Near but below the spectral attractor (0.09–0.10). The bridge maintains substantial connectivity.
- **Co/cross: 1.12:1.** Slight directional preference—the bridge favors the RD co-axial pairs by 12% on average. This is above unity but far below the 70,000:1 achieved by the standard architecture with the same pair specification.
- **Deviation: 0.35.** The per-layer offsets have grown substantially from their initialized values, indicating that individual layers differentiate from the master bridge. This deviation is roughly one-sixth the deviation observed in the collapse regime (2.12 for WL-001/R-001).
- **Val loss: 0.4009.** Marginally better than both the standard contrastive (0.4015) and spectral-only (0.4020) experiments, though the difference is within noise.

Interpretation. The emanation architecture creates a tension between global coherence and local adaptation. The master bridge receives the contrastive gradient directly, developing slight co-axial preference. But the per-layer offsets dilute this preference: each layer adapts its bridge to local gradient signals, and these local adaptations do not consistently reinforce the global topology. The master bridge carries a weak geometric signal (12% co-axial preference); the offsets fragment it.

The result is a bridge that is *coherent but not structured*: it maintains connectivity (Fiedler near the attractor), develops weak directional preference (co/cross slightly above unity), and allows per-layer differentiation (growing deviation)—without achieving the dramatic block-diagonal separation that requires per-bridge contrastive specificity.

This hierarchical coherence regime suggests that topology programming requires direct contrastive feedback to individual bridge matrices. Indirection through a shared master bridge preserves connectivity but loses the topological specificity needed for block-diagonal emergence. The emanation architecture may have value for other purposes—regularization, transfer, compositional adapters—but it does not serve as a topology programmer.

7.4 Topology Taxonomy: Four Regimes

The complete experimental programme produces four distinct dynamical regimes, cleanly separated with no intermediate or overlapping outcomes.

1. **Block-diagonal** (geometric contrastive). Polytope-derived pair specifications with per-bridge contrastive loss. Co/cross: 42,000–474,000:1. Fiedler: 10^{-5} – 10^{-4} . Experiments: O-001, H-ch6, Seed-43, T-001r2.
2. **Spectral attractor** (no contrastive). Spectral regularization alone, no pair specification. Co/cross: $\sim 1:1$. Fiedler: 0.084–0.102. Experiments: H-ch3, H-ch4, H-ch8, H-ch12.
3. **Hierarchical coherence** (emanation architecture). Contrastive loss on shared master bridge with per-layer offsets. Co/cross: 1.12:1. Fiedler: 0.084. Experiment: E-001.
4. **Connectivity collapse** (incoherent contrastive). Non-spatial pair specifications (random or algebraic). Co/cross: $\sim 10^{-5}:1$. Fiedler: $\sim 10^{-5}$. Deviation: ~ 2.1 . Experiments: WL-001, R-001.

Two features of this taxonomy are notable. First, the collapse regime (Regime 4) is distinct from the spectral attractor (Regime 3). The spectral attractor represents a stable equilibrium at Fiedler ≈ 0.09 ; the collapse regime represents a destructive interaction where the contrastive loss overwhelms the spectral loss and drives connectivity to zero. The spectral attractor is the bridge’s resting state; the collapse regime is the bridge’s failure mode under incoherent programming.

Second, Regimes 2 and 3 are structurally similar (both near Fiedler ≈ 0.09 , both with co/cross near unity) but mechanistically different. The spectral attractor achieves connectivity through spectral loss alone. Hierarchical coherence achieves similar connectivity through a different pathway—master bridge with contrastive loss—and develops weak directionality that the spectral attractor does not. The regimes are adjacent in metric space but distinct in mechanism.

8 Discussion

8.1 The Bridge as Programmable Substrate

Paper 3 framed the result as *discovery*: the Steersman discovers rhombic dodecahedral geometry in the bridge matrix. The present paper reframes this as *programming*: the Steersman programs whatever geometry it is told, provided the pair specification corresponds to a coherent spatial structure.

The evidence supports the reframing. The tesseract result (Section 4) demonstrates that the Steersman programs 4D geometry with the same fidelity as 3D: four independent 2×2 blocks, clean eigenvalue split, coupling ratio exceeding 41,000:1. The octahedral result (O-001) demonstrates programming at minimum dimensionality, achieving the programme’s strongest block-diagonal signal

(473,622:1 per-bridge mean). The spectral attractor (Section 5) shows that the bridge has a natural unprogrammed state—connected, isotropic, Fiedler ≈ 0.09 —and that contrastive loss adds direction to this connectivity.

The four-regime taxonomy (Section 6) sharpens the framing further. The bridge is a programmable substrate, but the programming language is *spatial*. The Steersman compiles polytope-derived pair specifications into block-diagonal structure. It rejects non-spatial specifications—random partitions and number-theoretic relationships—by driving the bridge into connectivity collapse. The selectivity is a feature: it confirms that successful topology programming encodes genuine geometric information, not an artifact of the loss formulation.

8.2 The Selective Geometric Compiler

The central thesis upgrade from Paper 3 is precise. The Steersman is not a universal topology enforcer—it does not program arbitrary pair specifications into block structure. It is a *selective geometric compiler*: it accepts pair specifications derived from the co-axial face or vertex structure of polytopes and rejects all others.

This selectivity is demonstrated by the four-way contrast at $n = 6$:

- **RD pairs (H-ch6):** 70,404:1. Spatial specification. Block-diagonal.
- **Emanation (E-001):** 1.12:1. Same spatial pairs, indirect application through master bridge. Weak coherence but no block-diagonal.
- **Wrong-labels (WL-001):** 8.7×10^{-6} :1. Random pairs. Connectivity collapse.
- **Resonance (R-001):** 8.9×10^{-6} :1. Number-theoretic pairs. Connectivity collapse.

The selectivity has two dimensions. The first is geometric: the pair specification must encode a coherent spatial decomposition (Regimes 1 vs. 4). The second is architectural: the contrastive signal must reach individual bridge matrices directly, not through hierarchical indirection (Regimes 1 vs. 2).

8.3 Number-Theoretic Structure Is Insufficient

The functional equivalence of WL-001 and R-001 (maximum divergence $< 10\%$ across 10,000 steps) has a clear implication: *mathematical structure without spatial embedding carries no information that the contrastive loss can use*. The Sophie Germain relationship ($2(11) + 1 = 23$), the twin prime pair (29, 31), and the residue class partition are genuine algebraic structures. They produce the same outcome as random noise.

This result constrains the class of viable topology specifications. The pair specification must encode a property of the bridge’s gradient landscape, not merely a mathematical relationship among channel indices. Polytope co-axial structure encodes such a property: it identifies axis-aligned subspaces of the bridge matrix that can be independently optimized. Number-theoretic relationships do not identify such subspaces.

8.4 Spectral-Contrastive Decomposition

The decomposition of the Steersman into spectral (connectivity) and contrastive (topology) components has a precise analogue in graph theory. The Fiedler eigenvalue measures algebraic connectivity [6]—how well-connected the graph is. The contrastive loss specifies edge weights—which

connections are strengthened. Spectral regularization without contrastive specification produces an Erdős–Rényi-like coupling pattern (all edges approximately equal); adding contrastive specification produces a stochastic block model [8] with known community structure.

The four regimes map onto this decomposition:

- **Spectral attractor:** Spectral loss alone. Connectivity without topology. The bridge graph is an approximately regular graph.
- **Block-diagonal:** Spectral + coherent contrastive. The bridge graph is a disjoint union of cliques (the blocks).
- **Hierarchical coherence:** Spectral + indirectly applied contrastive. The bridge graph is weakly modular—slight clustering without full community separation.
- **Collapse:** Spectral + incoherent contrastive. Destructive interference between the two losses drives the graph toward the empty graph.

This decomposition suggests that the two loss components are independently tunable but interact non-linearly. When they are aligned (coherent contrastive + spectral), they produce strong structure. When they conflict (incoherent contrastive + spectral), the contrastive loss dominates and suppresses connectivity below the spectral target. Understanding this interaction is a prerequisite for designing more sophisticated topology specifications.

8.5 The Inverse- n Relationship

The coupling ratio hierarchy—O-001 ($n = 4$, $\sim 474\text{K}:1$) $>$ H-ch6 ($n = 6$, $\sim 70\text{K}:1$) $>$ T-001r2 ($n = 8$, $\sim 42\text{K}:1$)—follows a clear inverse relationship with channel count. The mechanism is straightforward: at higher n , the bridge has more cross-axial entries to suppress, diluting the contrastive signal per entry. At $n = 4$, only 4 cross-axial pairs compete with 2 co-axial pairs; at $n = 8$, 24 cross-axial pairs compete with 4 co-axial pairs. The co-axial/cross-axial pair count ratio ($|C|/|\mathcal{X}|$) is $2/4 = 0.50$ for octahedral, $3/12 = 0.25$ for RD, and $4/24 = 0.167$ for tesseract.

This suggests a practical guideline: *use the minimum channel count that admits the desired topology*. Higher channel counts provide more topological expressiveness (more independent axes) but weaker per-axis signal. The practitioner’s choice of n trades off topological resolution against coupling strength.

8.6 Implications for Adapter Design

If the bridge is a programmable topology substrate, adapter design acquires a new degree of freedom. Current practice treats the rank dimension as homogeneous. Multi-channel extensions introduce structure but leave it to the optimizer to discover whatever coupling pattern the task prefers. The Steersman enables *prescribed* coupling: the practitioner specifies which channels should interact, and the feedback mechanism enforces the specification during training.

- **Topology as hyperparameter.** The pair specification is now a hyperparameter alongside rank, learning rate, and target modules. Different polytope geometries may suit different task families.
- **Compositional adapters.** Adapters with known block-diagonal structure compose more predictably than dense adapters: blocks can be independently swapped, scaled, or merged. The 2×2 block is the natural unit of composition.

- **Multi-modal bridges.** When adapting models across modalities (text, image, audio), the pair specification could encode known cross-modal relationships in the bridge structure.
- **Diagnostic topology.** Train with $n = 6$ or $n = 8$ and cybernetic feedback to discover the bridge’s preferred topology, then deploy with lower n for parameter efficiency. Paper 3 showed that $n = 3$ matches $n = 6$ task performance with $4\times$ fewer bridge parameters.

8.7 Limitations

Several limitations qualify the present results.

Single task family. All experiments use instruction-following fine-tuning on alpaca-cleaned. Whether the four-regime taxonomy, spectral attractor, and topology programming transfer to other task types (classification, summarization, code generation) remains untested.

Pair-based specifications only. The current pair specification interface partitions channels into pairs. More complex topologies—cliques of size > 2 , hierarchical nesting, cyclic structures—would require extending the contrastive loss formulation.

No formal optimality proof. We demonstrate that the Steersman programs a given topology, but we do not prove that any particular topology is optimal for any downstream task. The topology is a hyperparameter; its relationship to task performance remains to be characterized.

No theoretical account of the attractor. The spectral attractor (Fiedler ≈ 0.09) is empirically consistent across channel counts, but we provide no theoretical derivation of the equilibrium value. A theoretical account, possibly through random matrix theory or mean-field analysis of the spectral loss dynamics, is future work.

No theoretical account of geometric selectivity. We observe that only polytope-derived pair specifications produce block-diagonal structure, but we do not provide a formal characterization of which pair specifications constitute “valid geometric programs.” Understanding why the contrastive loss landscape has minima at polytope-aligned configurations and not at algebraically structured but spatially unembedded configurations is an open question.

Single seed for most experiments. All experiments use random seed 42 except Seed-43 and Seed-44 (which use seeds 43 and 44 respectively). While the binary regime classification is robust given the magnitude of the effects, specific numerical quantities—coupling ratios, Fiedler values, deviation—are point estimates.

Emanation architecture incompletely explored. E-001 tests one hierarchical design (master bridge + $1/\sqrt{l}$ -scaled offsets). The space of hierarchical bridge architectures is large; other designs may produce different outcomes. The third regime (hierarchical coherence) is characterized by a single experiment.

Scale. All topology experiments use TinyLlama-1.1B. Paper 3 established scale consistency for RD contrastive at 1.1B and 7B; the generalization of 4D programming, geometric selectivity, and the four-regime taxonomy to larger models is untested.

9 Conclusion

The bridge matrix in a multi-channel LoRA adapter is a programmable topology substrate. The Steersman—a cybernetic feedback mechanism comprising three closed-loop control laws—compiles polytope-derived pair specifications into block-diagonal structure with coupling ratios exceeding 40,000:1, preserving the property across 3D (rhombic dodecahedron, octahedron) and 4D (tesseract) geometries.

The mechanism is confirmed at three dimensionalities and four channel counts:

- **Octahedron** ($n = 4$, 3D): two 2×2 blocks, per-bridge mean coupling ratio **473,622:1**. The programme’s strongest signal.
- **Rhombic dodecahedron** ($n = 6$, 3D): three 2×2 blocks, coupling ratio **70,404:1**. Paper 3’s result, reproduced across seeds (Seed-43: 73,309:1).
- **Tesseract** ($n = 8$, 4D): four 2×2 blocks, coupling ratio **41,564:1**. Clean 4+4 eigenvalue split. Reproducible ($r = 1.0000$).

Three findings sharpen the understanding of this mechanism. *First*, the spectral attractor at Fiedler ≈ 0.09 is universal across channel counts ($n \in \{3, 4, 8, 12\}$, 10.4% band), providing the natural connectivity equilibrium from which contrastive programming departs. *Second*, the Steersman is a selective geometric compiler: it accepts spatial pair specifications and rejects algebraic ones, as demonstrated by the functional equivalence of wrong-label and resonance controls ($< 10\%$ trajectory divergence over 10,000 steps). *Third*, the inverse- n relationship—474K:1 ($n = 4$) $>$ 70K:1 ($n = 6$) $>$ 42K:1 ($n = 8$)—establishes a practical design principle: minimum geometry produces maximum signal.

The four-regime taxonomy—block-diagonal, spectral attractor, hierarchical coherence, and connectivity collapse—provides a complete map of the bridge’s dynamical behavior under different programming conditions. These regimes are cleanly separated with no intermediate outcomes, suggesting that the spectral-contrastive decomposition is a fundamental feature of the loss landscape.

Topology is now a hyperparameter. The choice of polytope geometry joins rank, learning rate, and target modules as a design decision for adapter construction. The pair specification is the program; the Steersman is the compiler; the bridge is the substrate. The compiler accepts valid geometric programs and rejects incoherent ones—and this selectivity confirms that successful topology programming encodes genuine geometric information, not an artifact of the loss formulation.

A Experimental Details

A.1 Model and Training Configuration

All experiments use TinyLlama-1.1B-Chat-v1.0 with the following hyperparameters held constant across all experiments:

Table 13: Shared training configuration across all experiments.

Parameter	Value
Base model	TinyLlama-1.1B-Chat-v1.0
Dataset	alpaca-cleaned [15]
Training steps	10,000
Learning rate	2×10^{-4}
LR scheduler	Cosine
Batch size	2
Gradient accumulation	8 (effective batch 16)
LoRA rank (r)	24
LoRA alpha	32
LoRA target modules	q-proj, k-proj, v-proj, o-proj
Initialization	Identity
Max sequence length	512
Precision	bfloat16
Random seed	42 (except Seed-43: 43, Seed-44: 44)

The channel size $s = r/n$ varies with channel count: $s = 8$ ($n = 3$), $s = 6$ ($n = 4$), $s = 4$ ($n = 6$), $s = 3$ ($n = 8$), $s = 2$ ($n = 12$). All experiments run to 10,000 steps with checkpoints saved every 100 steps for the first 1,000 steps and every 500 steps thereafter.

A.2 Steersman Configuration

The Steersman applies two loss components to the bridge matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$:

Spectral loss. Regularizes the Fiedler eigenvalue λ_2 of the weighted graph Laplacian $\mathbf{L} = \mathbf{D} - |\mathbf{M}|$, where $\mathbf{D}$ is the degree matrix of $|\mathbf{M}|$. This loss encourages algebraic connectivity without specifying direction.

Contrastive loss. Given a set of co-axial pairs $\mathcal{C}$ and the complementary set of cross-axial pairs $\mathcal{X}$, the contrastive loss rewards $|\mathbf{M}_{ij}|$ for $(i, j) \in \mathcal{C}$ and penalizes $|\mathbf{M}_{ij}|$ for $(i, j) \in \mathcal{X}$.

Control laws. The Steersman monitors three diagnostics: (1) connectivity trending (Fiedler eigenvalue trajectory), (2) directionality stagnation (co-axial/cross-axial coupling ratio plateau detection), and (3) stability monitoring (loss component variance). These three scalar diagnostics modulate the relative weighting of spectral and contrastive losses. The control laws are identical across all experiments; only the pair specification $\mathcal{C}$ changes.

A.3 Experiment Index

Table 14: Complete experiment index. All experiments use identical Steersman configuration and training hyperparameters; only n , pair specification, and architecture differ.

ID	Topology	n	$ \mathcal{C} $	Pairs	Architecture	Section
O-001	Octahedral	4	2	(0, 1), (2, 3)	Independent	6
H-ch6	RD contrastive	6	3	(0, 5), (1, 4), (2, 3)	Independent	6
Seed-43	RD contrastive	6	3	(0, 5), (1, 4), (2, 3)	Independent	6
Seed-44	RD contrastive	6	3	(0, 5), (1, 4), (2, 3)	Independent	6
T-001r2	Tesseract	8	4	(0, 1), (2, 3), (4, 5), (6, 7)	Independent	4
H-ch3	Spectral-only	3	—	—	Independent	5
H-ch4	Spectral-only	4	—	—	Independent	5
H-ch8	Spectral-only	8	—	—	Independent	5
H-ch12	Spectral-only	12	—	—	Independent	5
E-001	Emanation	6	3	(0, 5), (1, 4), (2, 3)	Hierarchical	7
WL-001	Wrong-labels	6	3	Random partition	Independent	6
R-001	Resonance	6	3	(0, 2), (3, 4), (1, 5)	Independent	7

A.4 Metrics

Co-axial/cross-axial coupling ratio (ρ). For each bridge matrix $\mathbf{M}$, compute the mean absolute value of co-axial entries (those at positions $(i, j) \in \mathcal{C}$) and cross-axial entries (those at positions $(i, j) \in \mathcal{X}$). The ratio $\rho = |\bar{M}_{\mathcal{C}}| / |\bar{M}_{\mathcal{X}}|$ measures directional preference. Values $\gg 1$ indicate block-diagonal structure; values ≈ 1 indicate isotropy; values $\ll 1$ indicate suppression of the specified co-axial direction.

Fiedler eigenvalue (λ_2). The second-smallest eigenvalue of the weighted graph Laplacian $\mathbf{L} = \mathbf{D} - |\mathbf{M}|$. Measures algebraic connectivity of the bridge. Values near zero indicate near-disconnection (block-diagonal or collapse); values near 0.1 indicate stable connectivity (spectral attractor).

Deviation from identity. Mean Frobenius norm of the difference between each bridge matrix and the identity matrix, normalized by dimension. Measures how far the bridge has departed from initialization. Reported primarily for the collapse and emanation regimes.

Eigenvalue split. For n -channel bridges with k co-axial pairs, successful topology programming produces a $k + (n - k)$ eigenvalue split: k eigenvalues elevated (within-block connectivity) and $n - k$ eigenvalues suppressed to $\mathcal{O}(10^{-4})$ (between-block disconnection).

A.5 Hardware

Experiments were conducted on two machines:

- **Local:** NVIDIA RTX 6000 Ada Generation (48 GB VRAM). Used for T-001r2 (tesseract contrastive).
- **Hermes:** NVIDIA RTX 4090 (16 GB VRAM). Used for spectral-only ablation (H-ch3, H-ch4, H-ch8, H-ch12), wrong-labels (WL-001), resonance (R-001), emanation (E-001), octahedral (O-001), and seed replication (Seed-43, Seed-44).

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